

Hands-on Lab: Generative AI for Data Generation and Augmentation

Estimated time needed: 30 minutes

Introduction

One of the principal advantages of generative AI is its ability to generate realistic synthetic data. Synthetic data is generated when a pretrained generative model responds to a prompt, creates new data samples, or performs transfer learning on a given dataset. In addition, it generates samples that augment the existing data set while preserving its statistical distribution and interpretability.

In this lab, you will learn how to use generative AI to generate synthetic data samples and transfer learning on a given dataset.

Learning objectives

By the end of this lab, you will be able to understand and apply key concepts of synthetic data generation using generative AI tools in practical scenarios.

- Understand the concept of synthetic data generation using generative AI
- Use tools such as **Tonic Fabricate** and **ilovecsv** to create synthetic data samples
- Augment an existing CSV dataset while maintaining its statistical characteristics
- Explore how generative AI can enhance datasets for analysis, modeling, and other downstream tasks

Insurance dataset overview

The dataset used in this lab includes insurance records.

This dataset is a cleaned version of the original dataset, meaning the data has been processed to remove errors, handle missing values, and ensure consistency.

The original dataset, [Medical Insurance Price Prediction](#), contains raw, unprocessed data and is available on [Kaggle](#) under the [CC0 1.0 Universal License](#), which allows free use and distribution.

You can download the cleaned dataset file directly from the link provided here:

[Insurance Dataset](#)

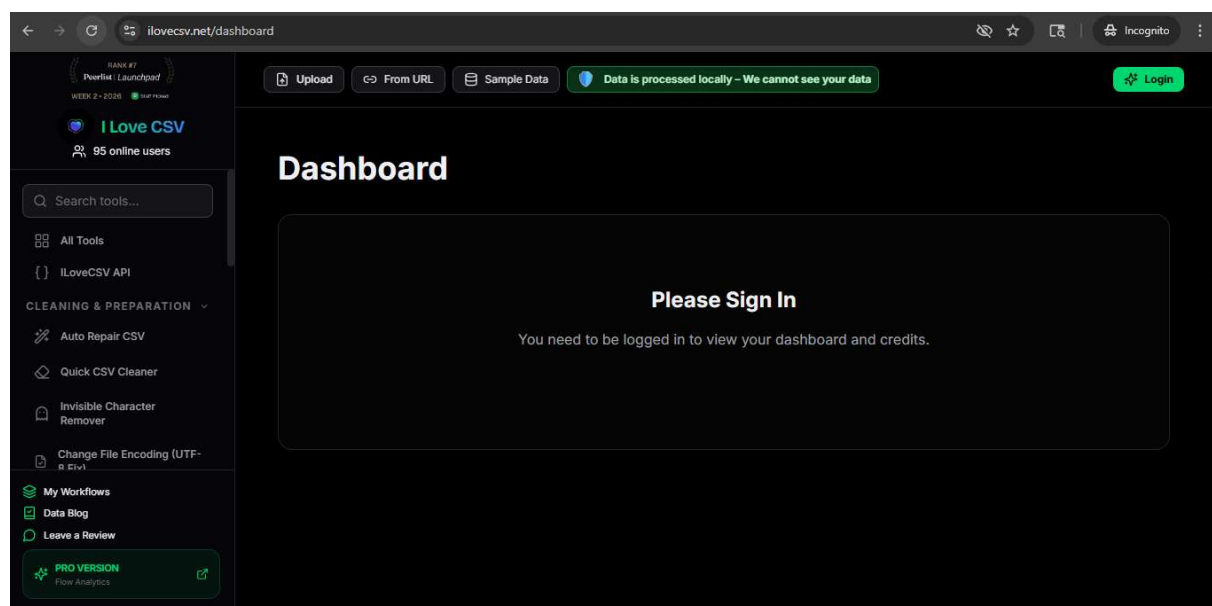
Option 1: ilovecsv Tool

The **ilovecsv** tool is a simple, GUI-based tool. You will now explore how to use it.

1. Open the website

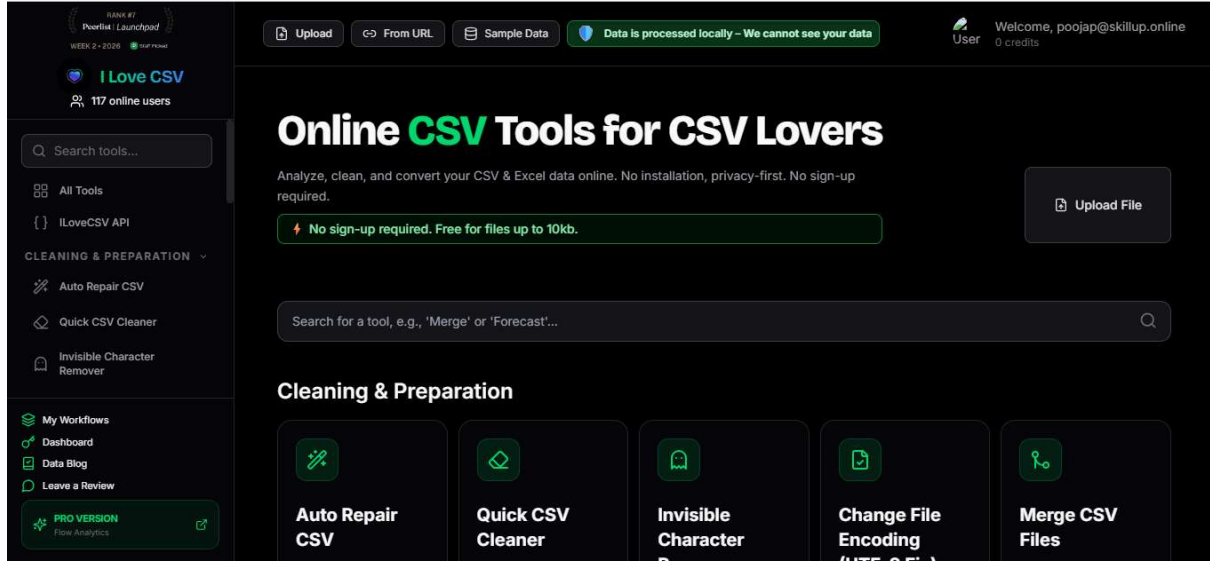
Click this [ilovecsv](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



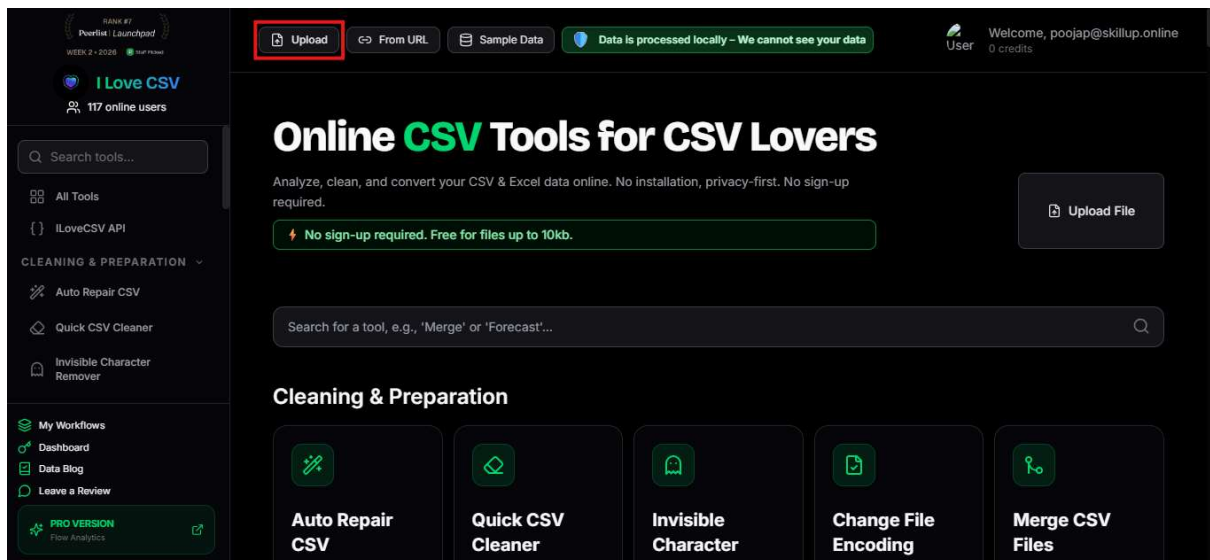
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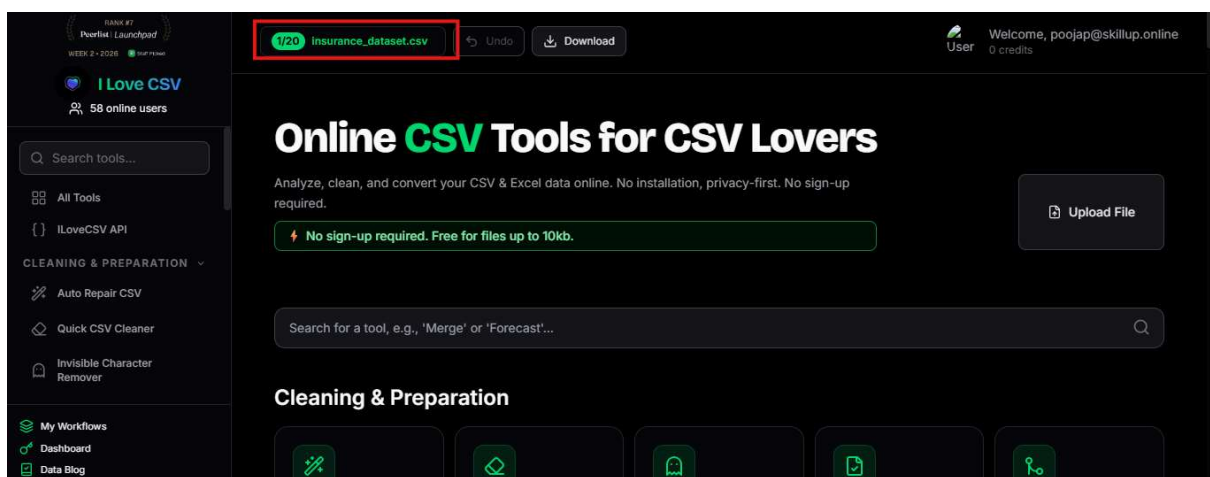


3. Upload the dataset

- Click on the **Upload** button given at the top of the page, and upload the CSV file of the data set to the interface.

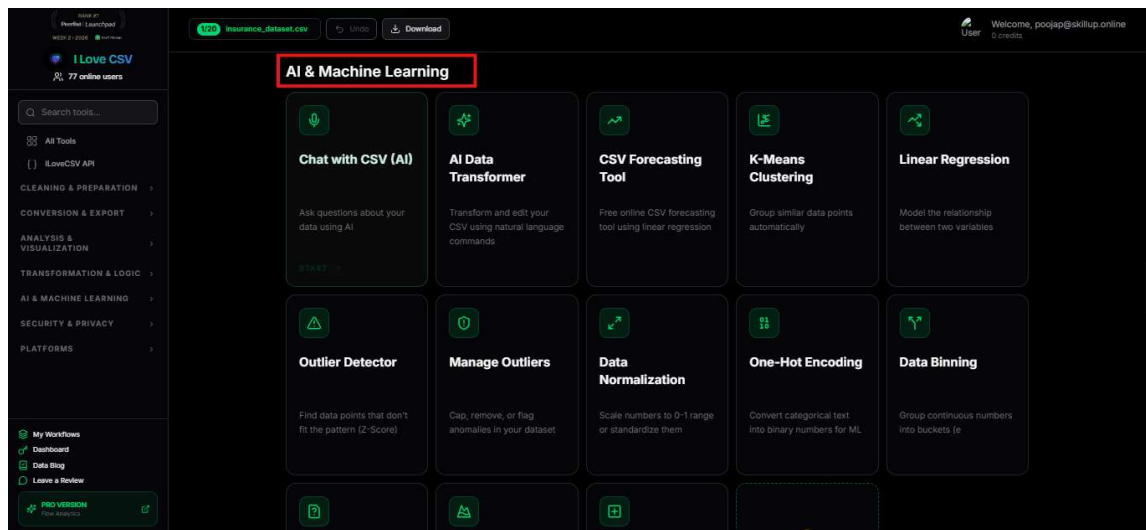


- After uploading the dataset, you will be able to see the **insurance_dataset** file listed in the uploaded files section.

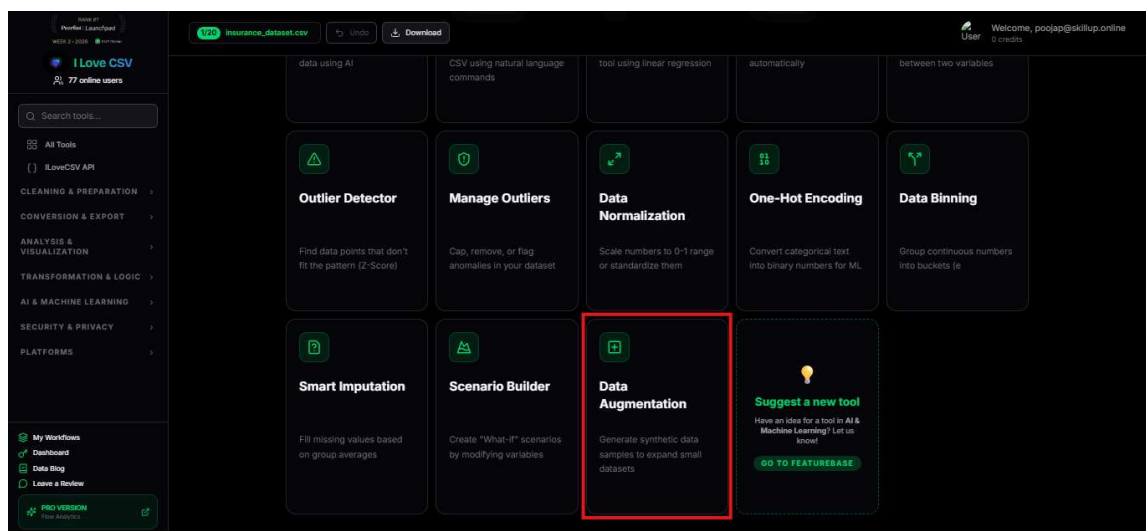


4. Access the Data Augmentation tool

- You are on the main dashboard page. Scroll down through the available sections until you locate the **AI and Machine Learning** section, or use the left-hand navigation panel to quickly navigate to it directly.



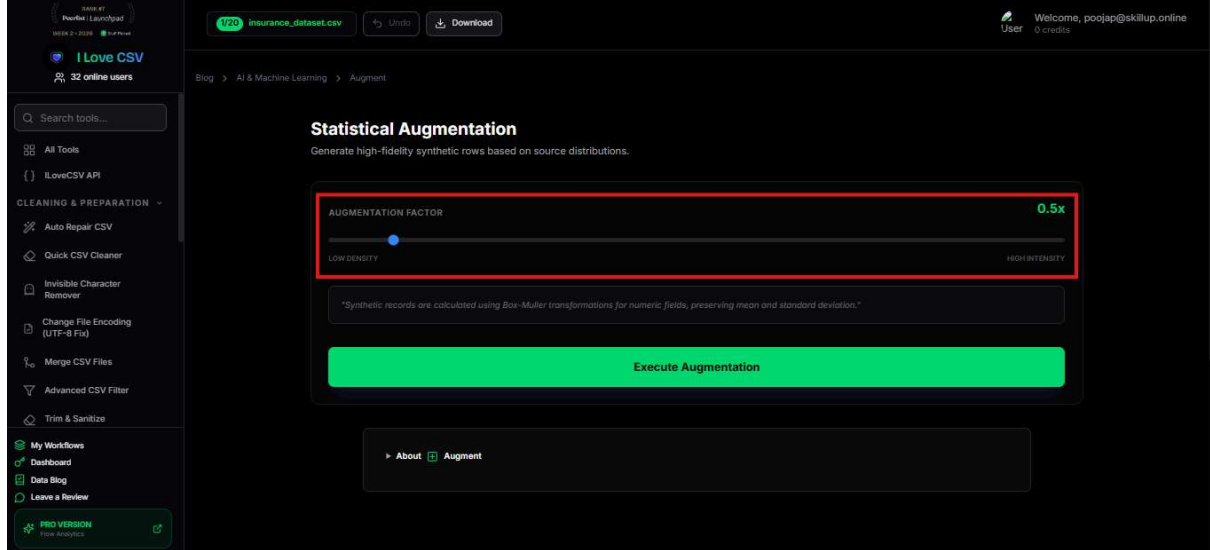
- Within the AI and Machine Learning section, locate the **Data Augmentation Tool** from the available options and click on it to open the tool interface.



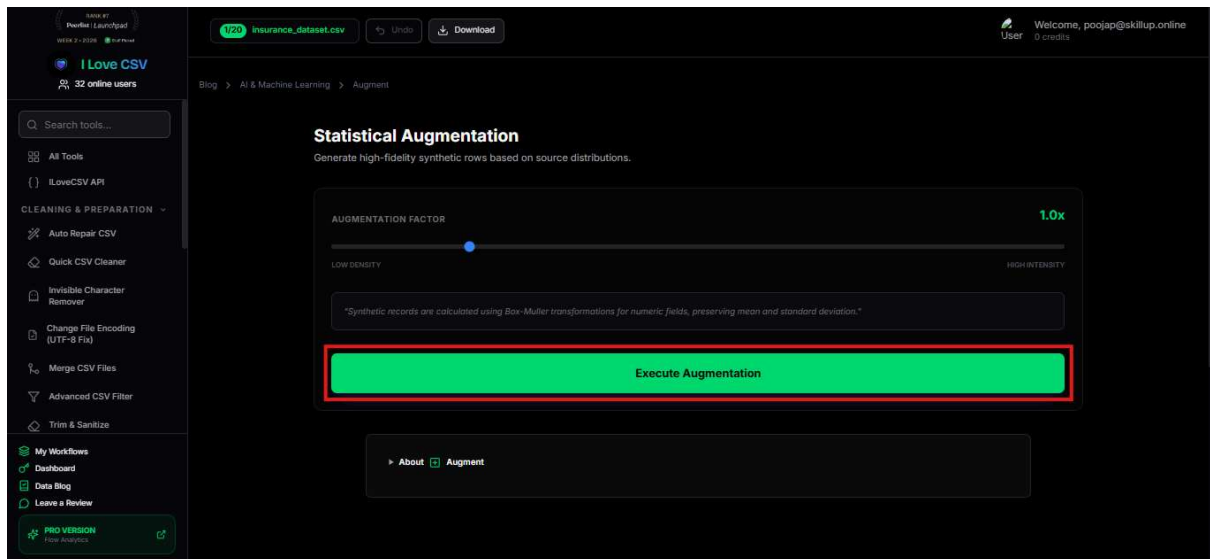
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Statistical augmentation in the ilovecsv tool involves generating synthetic rows from your existing dataset while preserving its statistical properties.

You can use the **Augmentation factor** to adjust the **density (number of rows)** of the synthetic dataset as required.

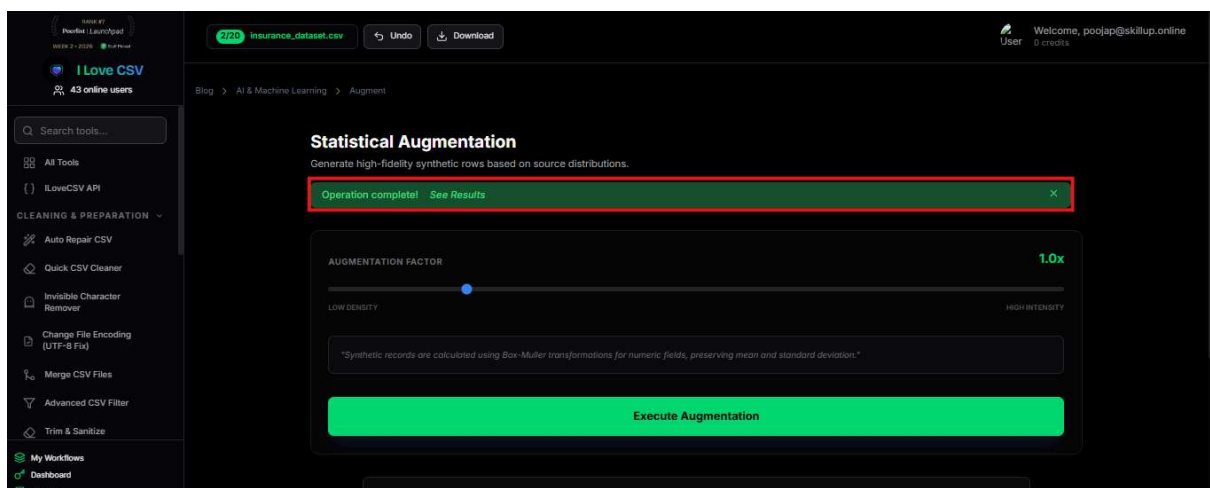


Once you have adjusted the augmentation factor (for example, 1x), click the **Execute Augmentation** button.

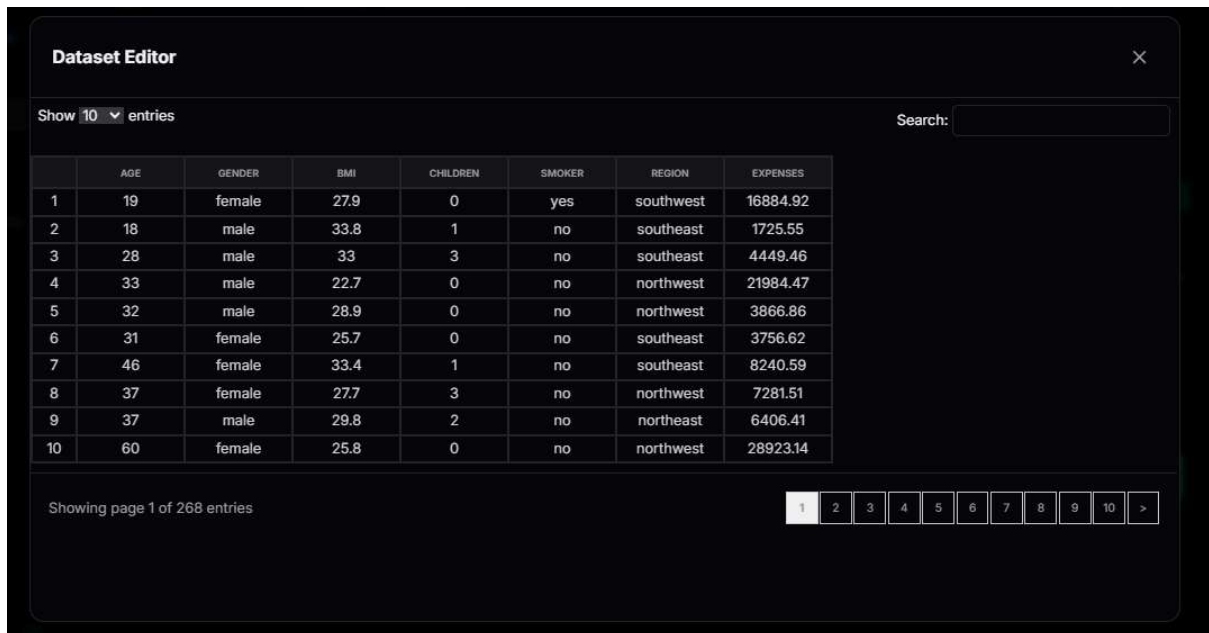


6. Download the synthetic data

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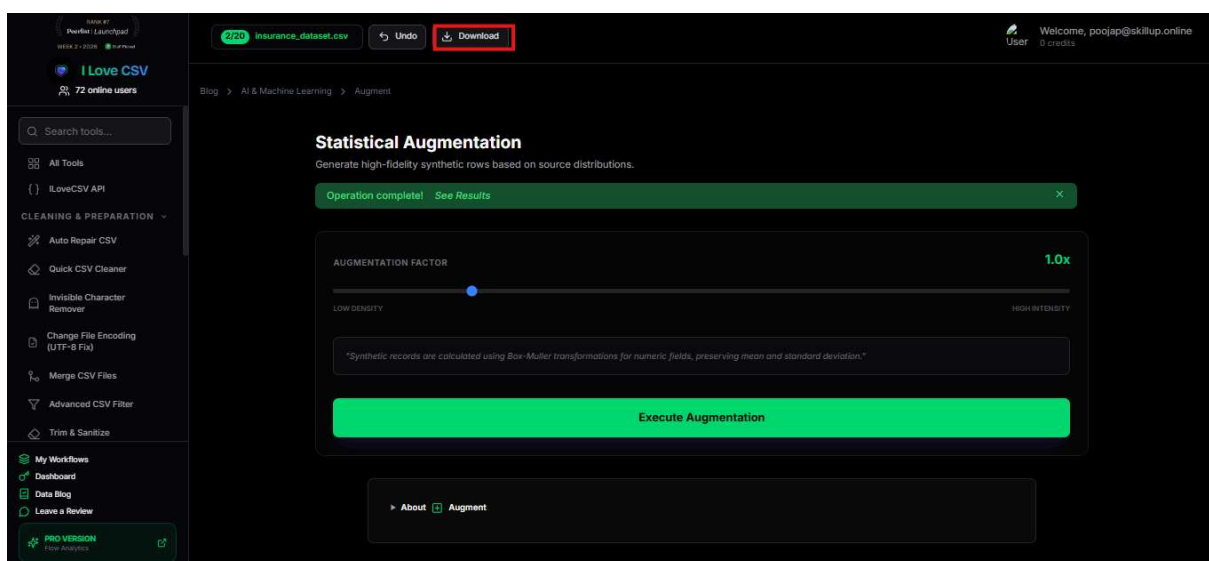
Click the [See Results](#) hyperlink to open and view the final results in the editor.



The screenshot shows a 'Dataset Editor' window with a table of 10 entries. The table has columns for AGE, GENDER, BMI, CHILDREN, SMOKER, REGION, and EXPENSES. Below the table, it indicates 'Showing page 1 of 268 entries' and a pagination control with buttons for 1 through 10 and a next button.

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3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.47
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9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.14

Close the results window and return to the previous page. Then click [Download](#) to save the output file to your local machine.



The screenshot shows the 'I Love CSV' web application interface. The main area displays the 'Statistical Augmentation' tool, which generates high-fidelity synthetic rows based on source distributions. The 'Augmentation Factor' is set to 1.0x. A green button labeled 'Execute Augmentation' is visible. The left sidebar contains various tool categories like 'CLEANING & PREPARATION' and 'My Workflows'. The top bar shows the current file 'insurance_dataset.csv' and a 'Download' button.

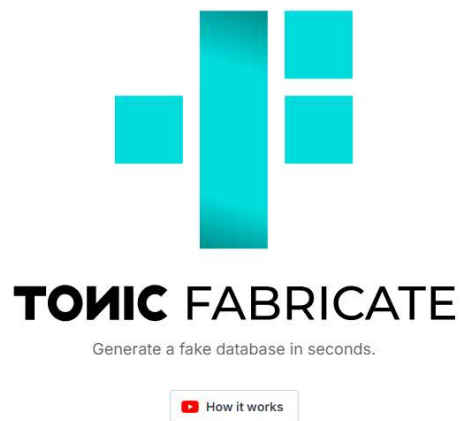
In the final result, you will get 2,676 rows, which is double the original file since you selected a 1× augmentation factor. You can now use this synthetic data set for data science operations, or you can also augment the original data set with these samples.

Option 2: Tonic Fabricate

1. Open the website

Click this [Tonic Fabricate](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:

The image shows a 'Create My Account' form on a dark background. At the top, the title 'Create My Account' is in white. Below the title are four circular icons for Google, GitHub, Microsoft, and a key icon. A line of text says 'or use your email for registration'. Below this are three input fields: 'Name', 'Email', and 'Password'. The 'Password' field has a note '6 characters minimum' below it. At the bottom, there is a small paragraph of text: 'Signing up signifies that you are authorized to enter into an agreement with Tonic, and that you have read and agree to our Terms of Use and Privacy Policy, which will govern all uses of any Tonic product, unless there is a signed agreement between you and Tonic that you agree governs this use.'

2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

The image shows the main interface of the Tonic Fabricate website. At the top left is the 'TONIC FABRICATE' logo. In the center, there is a heading 'What data can I fabricate for you today?'. Below the heading is a large text input box with the placeholder text 'Drop in a schema file or describe the type of data you want to get started.' and a dropdown menu labeled 'Agent'. Below the input box are three buttons: 'Generate data for an online store', 'Simulate bank transactions', and 'Create social media data'.

3. Data Augmentation

This is a prompt-based tool that lets you customize synthetic data to meet your data science operational requirements. Upload the Insurance dataset using the **Attach a File** option and copy and paste the below prompt in the message box.

The image shows a prompt input box. At the top left is a code icon. At the top right is a purple button labeled 'plaintext' and a copy icon. Below the header is a text input area with the prompt 'Generate a synthetic dataset based on the uploaded insurance data file. Pr'. The input area has a light blue background and a white border.

Note: Synthetic data generation in this tool may take some time, so please be patient during the process.

TONIC FABRICATE

What data can I fabricate for you today?

relationships between variables (e.g., higher age may correlate with higher premiums, and claim amounts should align with policy coverage). Avoid duplicates and unrealistic values. Generate clean, high-quality data suitable for analysis, keeping the synthetic data statistically consistent with the source dataset with approximately 80% similarity/accuracy. Create a dataset that is 1× the size of the original while introducing meaningful variations.

insurance_dataset.csv (50.2 KB) X

Agent

Generate data for an online store Simulate bank transactions

Create social media data

Click **Send** to submit the prompt to initiate the process.

TONIC FABRICATE

insurance_source 1338 records

Default SQL Memory Import Export

id	age	gender	bmi	children	smoker	region	expens
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	19	female	27.9	0	yes	southwest	16884.9
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9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.1
11	25	male	26.2	0	no	northeast	2721.32
12	62	female	26.3	0	yes	southeast	27808.7
13	23	male	34.4	0	no	southwest	1826.84
14	56	female	39.8	0	no	southeast	11090.7
15	27	male	42.1	0	yes	southeast	39611.7

Previous Page 1 of 14 Next

Synthetic Insurance Data Generation

Children 0-5, heavily skewed toward 0 (43%)

Smoker ~80% no, ~20% yes

Region ~24-27% per region

Expenses Strongly correlated with smoker status + age; smokers avg ~\$32K non-smokers avg ~\$8.4K

Now I'll create the target table and generate 1,338 synthetic rows preserving these distributions and correlations:

Create synthetic insurance table 6s

Generate synthetic insurance data 24s

All 1,338 synthetic rows generated. Let me now validate the quality by comparing distributions between source and synthetic data:

Ran query source vs synthetic comparison

Ran query region distribution comparison

Ran query expense patterns by age & smoker

Working...

Agent

4. Download the synthetic data

Once the process is complete, you will be prompted with the message: **The data is ready for export or analysis. Would you like me to download it as a CSV?.** Type **Yes** to proceed with downloading the dataset.

insurance 1338 records

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
4	38	female	24.7	1	no	southwest	2868.68
5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
15	49	male	31.2	1	no	southwest	11600.3

synthetic_insurance_dataset.csv
text/csv • 49.09 KB

Download

Here's your synthetic insurance dataset as a CSV (49 KB, 1,338 rows). The `id` column has been excluded to keep the format consistent with your original file.

What would you like to do next?

Agent

Click on **Download** to save the data to your local machine.

5. Analyze the report

This tool allows you to view details about the model used to generate the synthetic data.

Copy and paste the below prompt to see the report

plaintext

Provide a detailed report on the model used for generating the synthetic d

After you give the prompt, you will see a detailed report along with visualizations.

insurance 1338 records

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Synthetic Insurance Data Generator

Age Group	Smoker	Mean Expense	Max Expense	Similarity
60-64	No	\$15,233	\$16,094	5.7%
60-64	Yes	\$40,631	\$42,178	3.8%

The monotonic increase in expenses with age is preserved for both smoker groups. Smokers consistently cost ~3-4x more than non-smokers across all age buckets.

8. Summary Scorecard

Evaluation Dimension	Weight	Score
Numeric Distribution (KS Tests)	30%	96.8%
Categorical Distribution (TVD)	20%	99.0%
Correlation Preservation	25%	96.3%
Discriminator AUC	25%	98.4%
Overall Similarity Score	100%	97.5%

The synthetic dataset exceeds the 80% similarity target across all dimensions, scoring 97.5% overall. The data is statistically consistent with the source, preserves all key relationships, and is indistinguishable from the original by a machine learning classifier.

What would you like to do next?

Agent

Conclusion

Congratulations! You have completed the lab on data augmentation using iLoveCSV and Tonic Fabricate tools.

Author(s)

Pooja Patel



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Introduction

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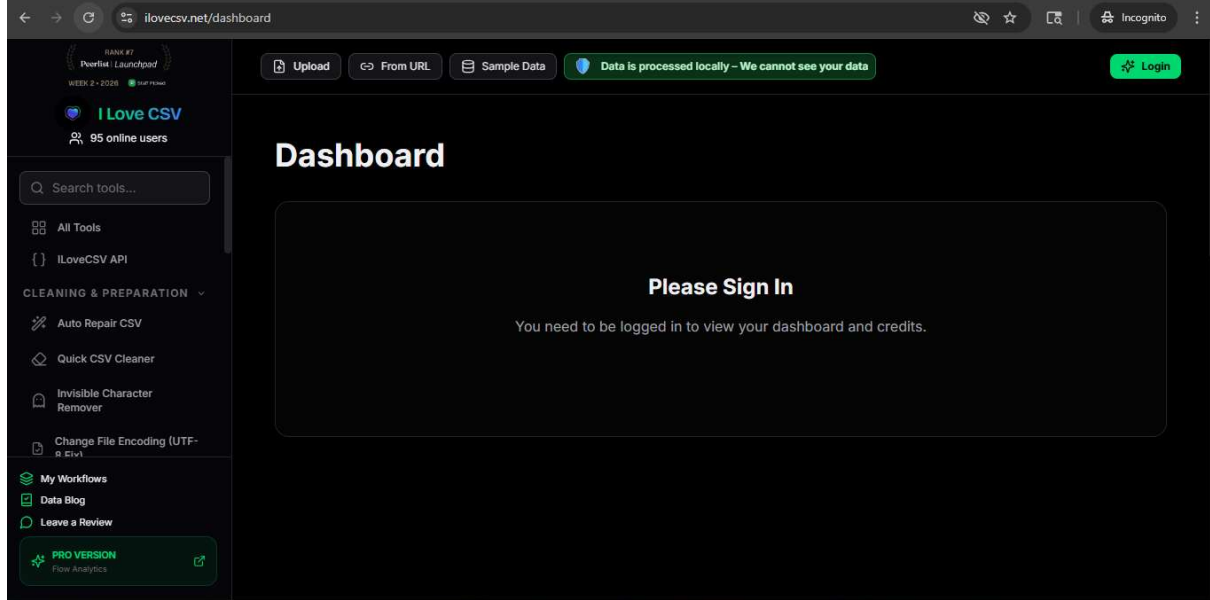
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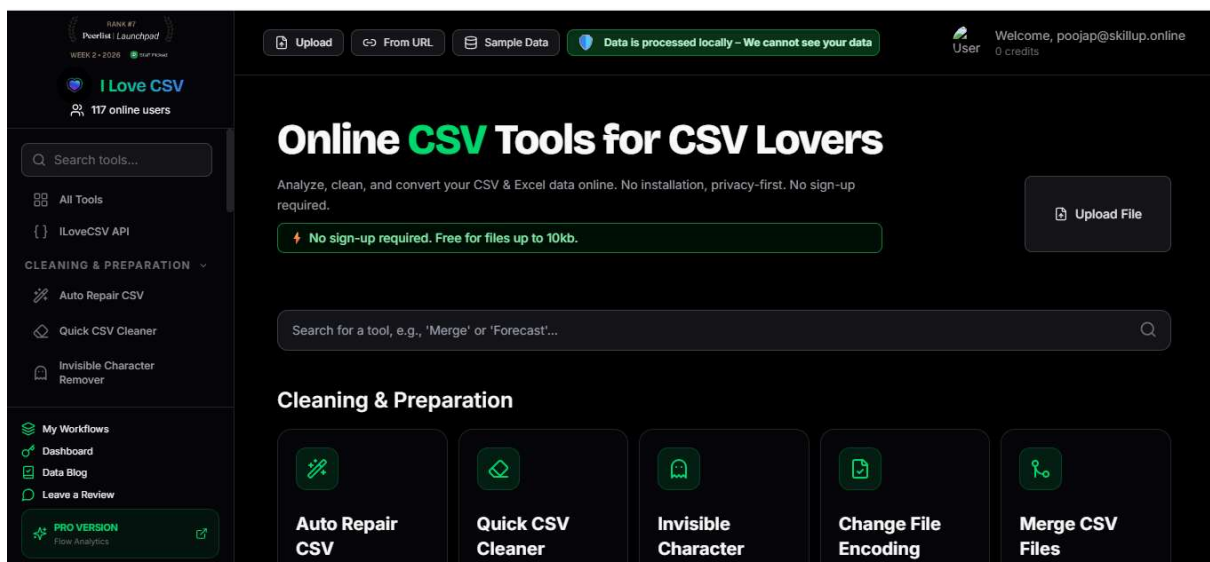
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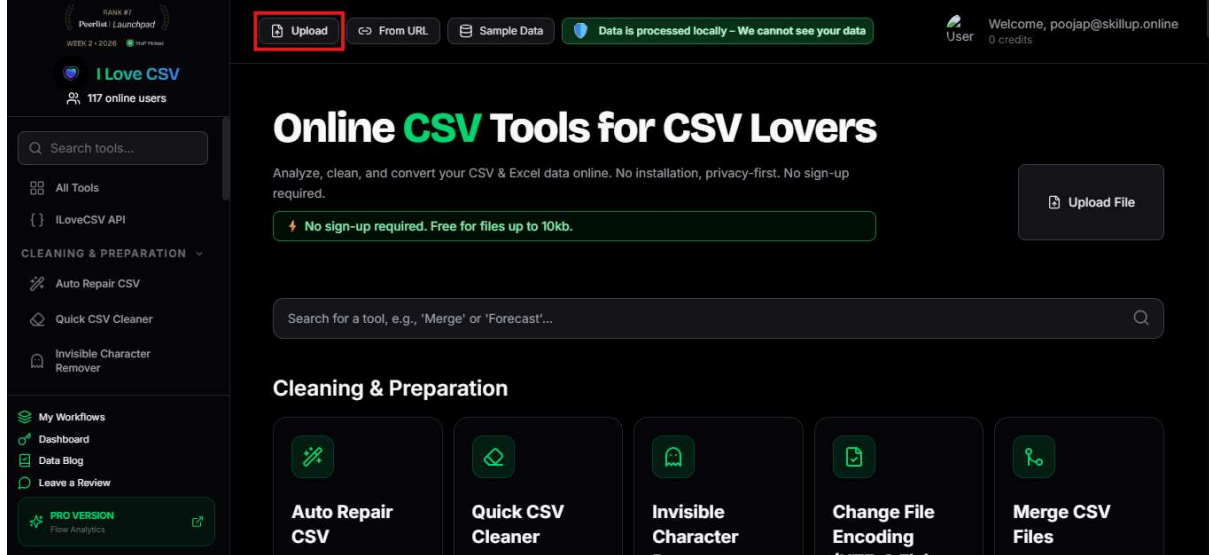
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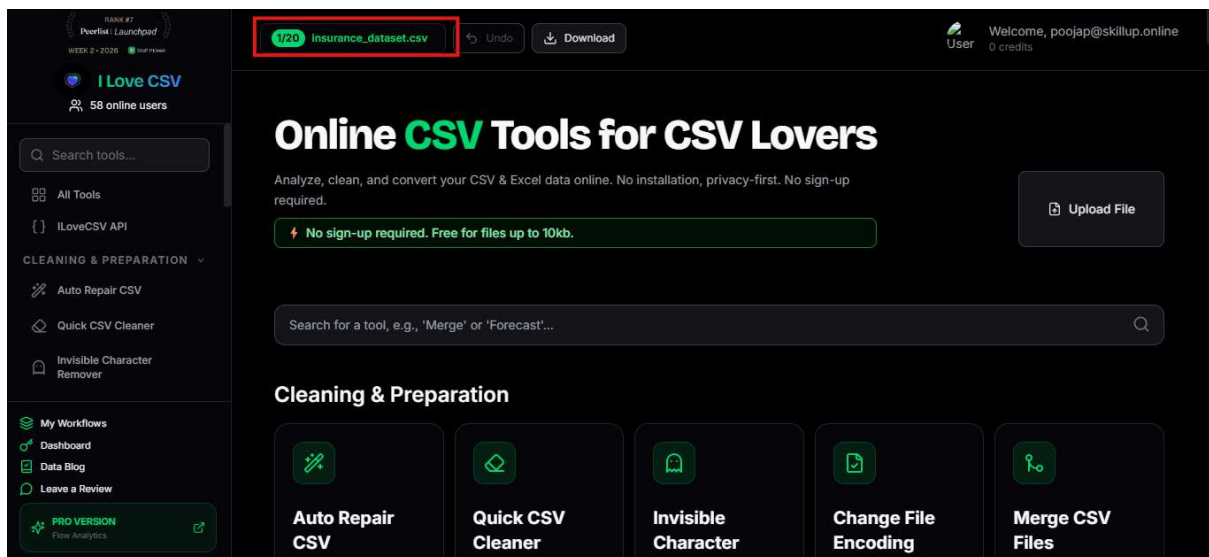


3. Upload the dataset

- Click on the **Upload** button given at the top of the page, and upload the CSV file of the data set to the interface.

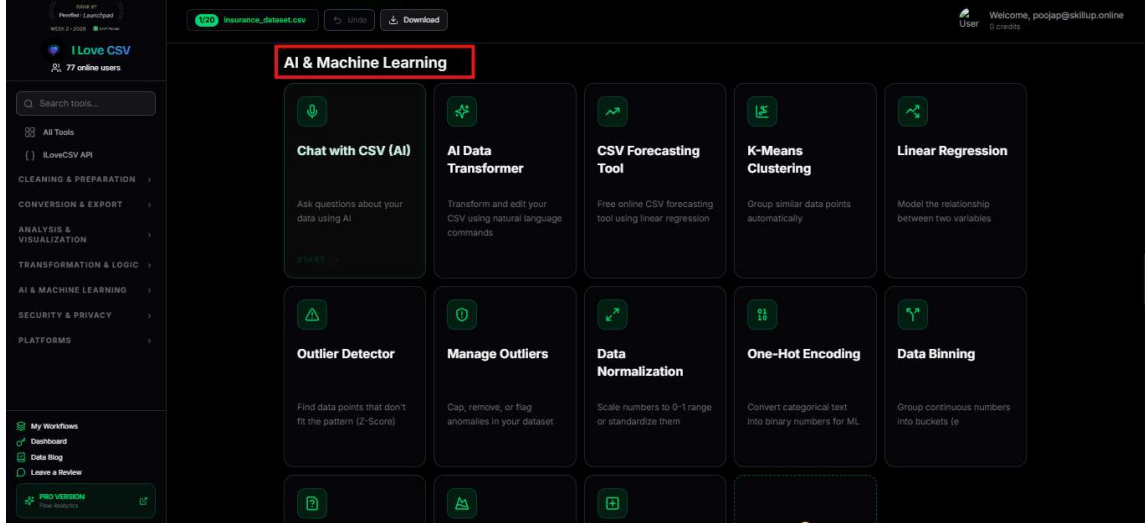


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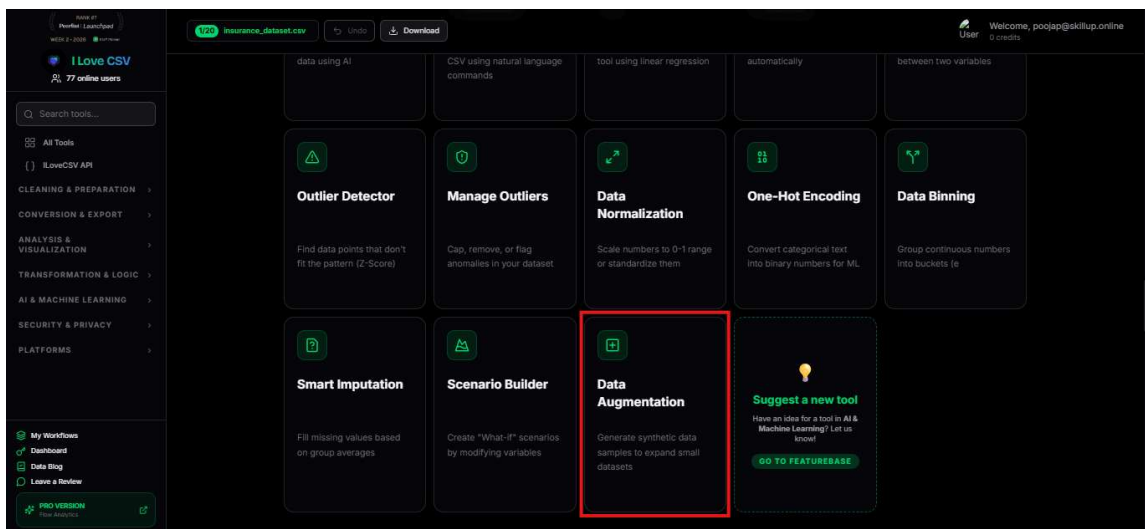


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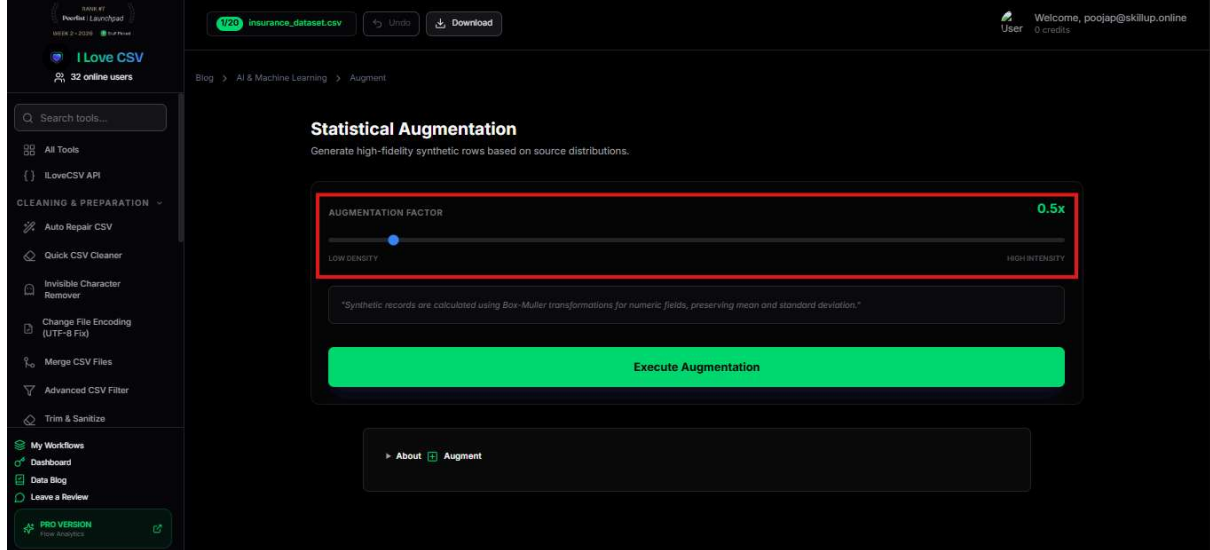
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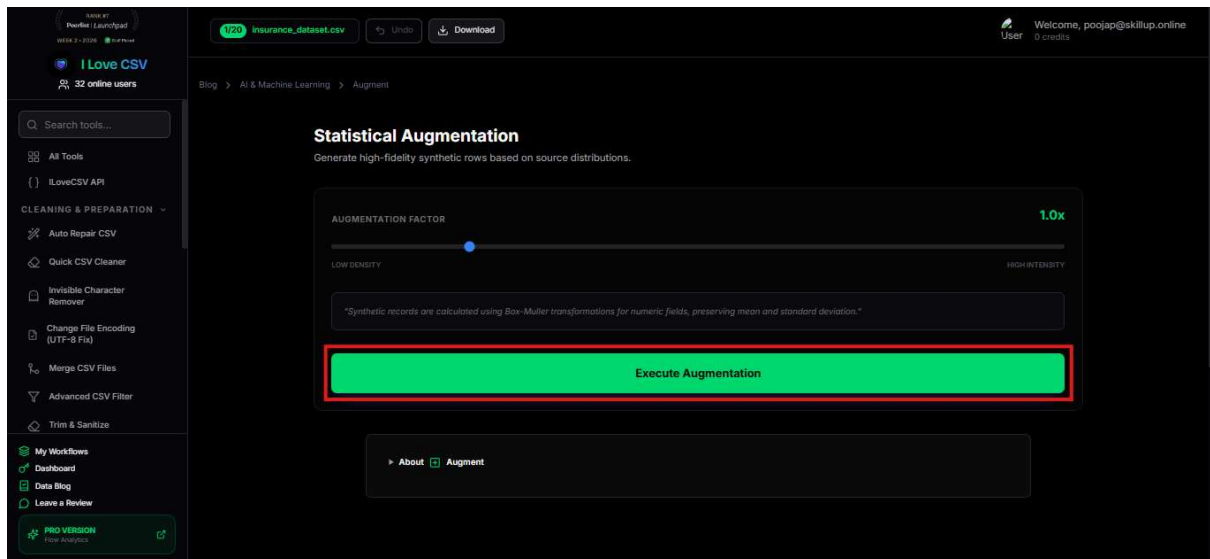
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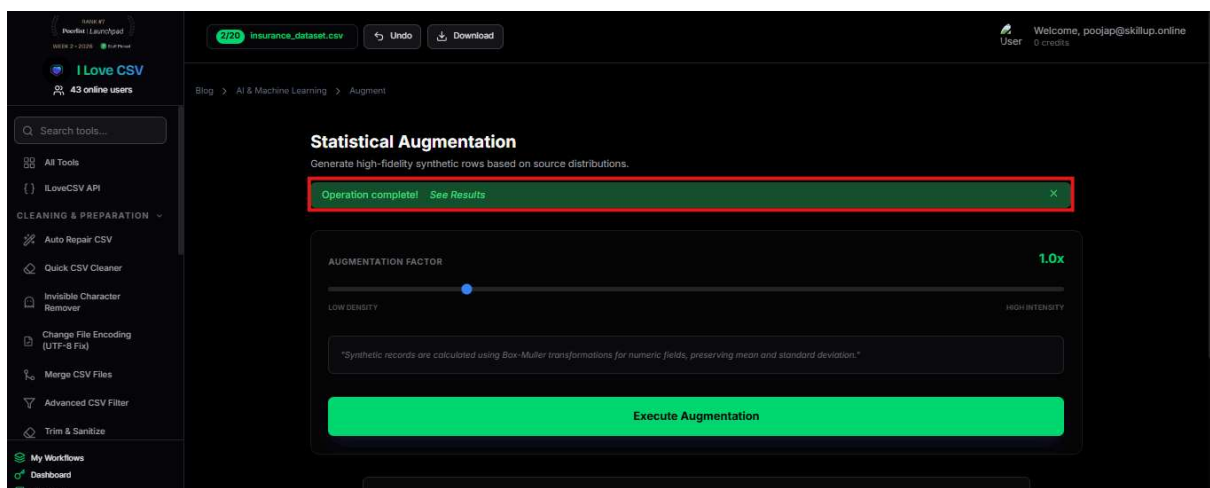


Once you have adjusted the augmentation factor (for example, 1x), click the **Execute Augmentation** button.

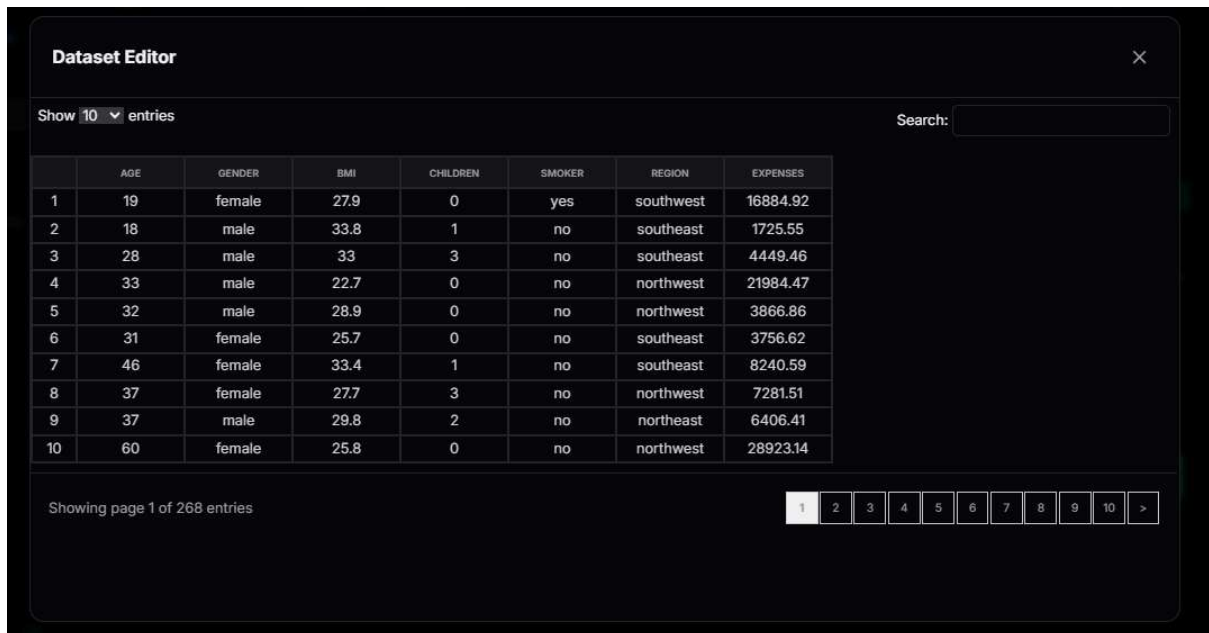


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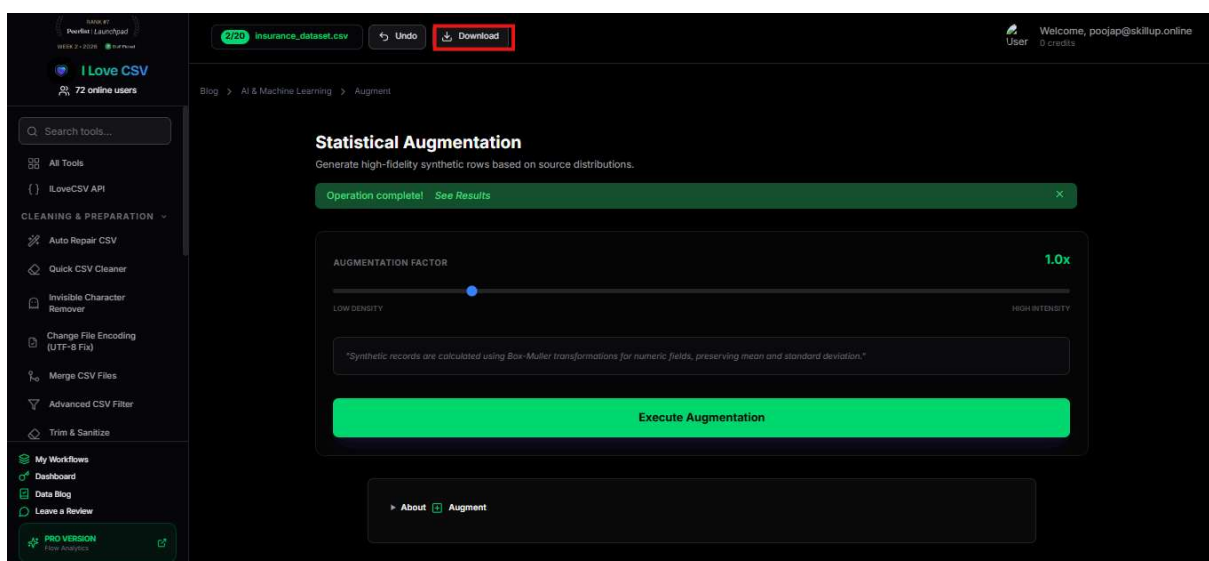
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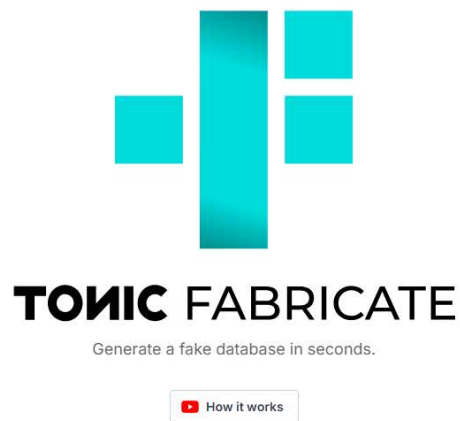
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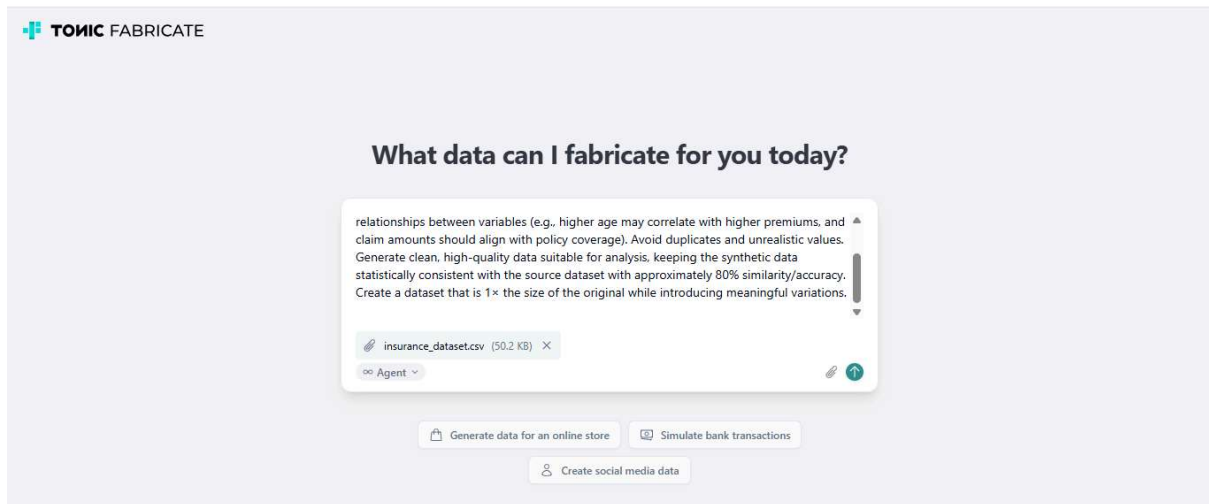
The image shows the main interface of the Tonic Fabricate website. At the top left is the 'TONIC FABRICATE' logo. In the center, there is a large text box with the prompt 'What data can I fabricate for you today?'. Below this prompt is a smaller text box with the instruction 'Drop in a schema file or describe the type of data you want to get started.' and a dropdown menu labeled 'Agent'. At the bottom, there are three buttons: 'Generate data for an online store', 'Simulate bank transactions', and 'Create social media data'.

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This is a prompt-based tool that lets you customize synthetic data to meet your data science operational requirements. Upload the Insurance dataset using the **Attach a File** option and copy and paste the below prompt in the message box.

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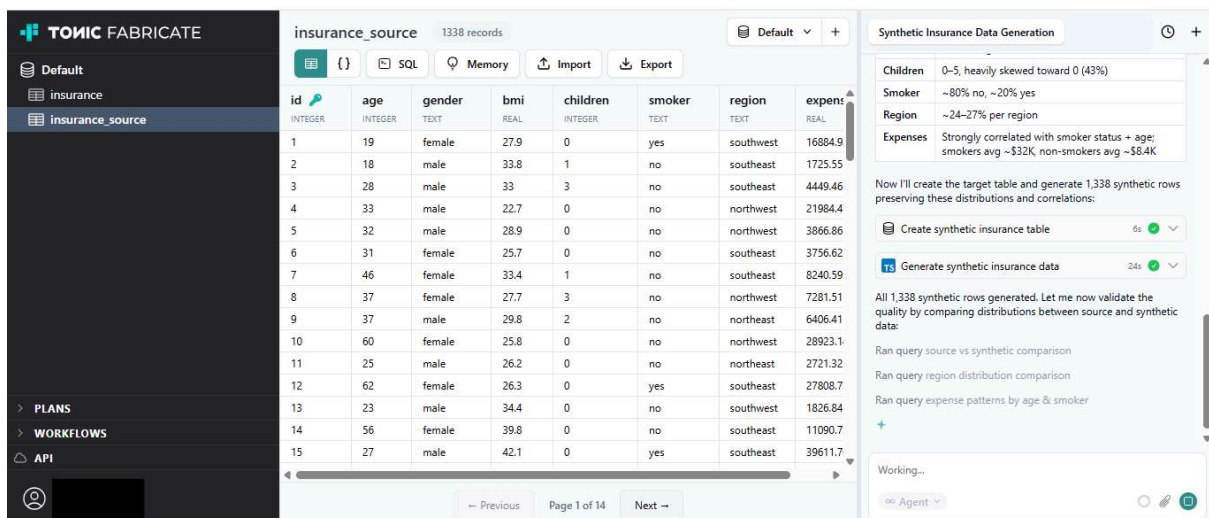
insurance_dataset.csv (50.2 KB) X

Agent

Generate data for an online store Simulate bank transactions

Create social media data

Click **Send** to submit the prompt to initiate the process.



TONIC FABRICATE

Default

insurance

insurance_source

PLANS

WORKFLOWS

API

insurance_source 1338 records

Default +

SQL Memory Import Export

id	age	gender	bmi	children	smoker	region	expens
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	19	female	27.9	0	yes	southwest	16884.9
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.4
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.1
11	25	male	26.2	0	no	northeast	2721.32
12	62	female	26.3	0	yes	southeast	27808.7
13	23	male	34.4	0	no	southwest	1826.84
14	56	female	39.8	0	no	southeast	11090.7
15	27	male	42.1	0	yes	southeast	39611.7

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Synthetic Insurance Data Generation

Children 0-5, heavily skewed toward 0 (43%)

Smoker ~80% no, ~20% yes

Region ~24-27% per region

Expenses Strongly correlated with smoker status + age; smokers avg ~\$32K non-smokers avg ~\$8.4K

Now I'll create the target table and generate 1,338 synthetic rows preserving these distributions and correlations:

Create synthetic insurance table 6s

Generate synthetic insurance data 24s

All 1,338 synthetic rows generated. Let me now validate the quality by comparing distributions between source and synthetic data:

Ran query source vs synthetic comparison

Ran query region distribution comparison

Ran query expense patterns by age & smoker

Working...

Agent

4. Download the synthetic data

Once the process is complete, you will be prompted with the message: **The data is ready for export or analysis. Would you like me to download it as a CSV?.** Type **Yes** to proceed with downloading the dataset.

insurance 1338 records

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
4	38	female	24.7	1	no	southwest	2868.68
5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
15	49	male	31.2	1	no	southwest	11600.3

synthetic_insurance_dataset.csv
text/csv • 49.09 KB

Download

Here's your synthetic insurance dataset as a CSV (49 KB, 1,338 rows). The `id` column has been excluded to keep the format consistent with your original file.

What would you like to do next?

Agent

Click on **Download** to save the data to your local machine.

5. Analyze the report

This tool allows you to view details about the model used to generate the synthetic data.

Copy and paste the below prompt to see the report

plaintext

Provide a detailed report on the model used for generating the synthetic d

After you give the prompt, you will see a detailed report along with visualizations.

insurance 1338 records

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
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9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
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What would you like to do next?

Agent

60-64 No \$15,233 \$16,094 5.7%

60-64 Yes \$40,631 \$42,178 3.8%

The monotonic increase in expenses with age is preserved for both smoker groups. Smokers consistently cost ~3-4x more than non-smokers across all age buckets.

8. Summary Scorecard

Evaluation Dimension	Weight	Score
Numeric Distribution (KS Tests)	30%	96.8%
Categorical Distribution (TVD)	20%	99.0%
Correlation Preservation	25%	96.3%
Discriminator AUC	25%	98.4%
Overall Similarity Score	100%	97.5%

The synthetic dataset exceeds the 80% similarity target across all dimensions, scoring 97.5% overall. The data is statistically consistent with the source, preserves all key relationships, and is indistinguishable from the original by a machine learning classifier.

What would you like to do next?

Agent

Conclusion

Congratulations! You have completed the lab on data augmentation using iLoveCSV and Tonic Fabricate tools.

Author(s)

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